

USBC-I2C

USB-C → 3.3 V → I²C breakout board

3.3V

OUTPUT

800mA

MAX CURRENT

50×30

MM

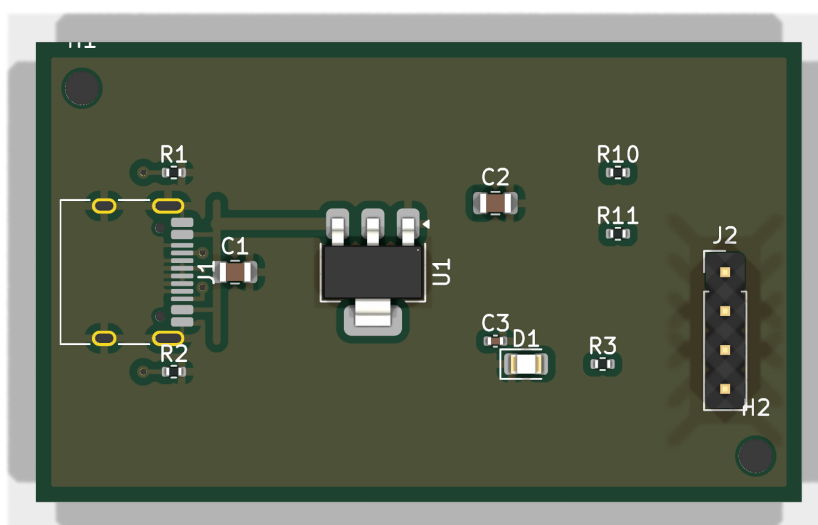


Figure 1. Board top view (3D render).

This document was generated by [circuit_toolkit](#) from a 25-line Python topology description. Any drift between schematic, PCB, and BOM is by construction impossible — they share a single source.

1.0 SPECIFICATIONS

Electrical and mechanical parameters derived from the circuit topology and PCB layout. Manufacturer-rated values reference the AMS1117 datasheet and USB Type-C specification.

Parameter	Value
Input	USB-C, 5V nominal (4.75–5.25V)
USB advertised role	Sink, 5V/3A (Rd = 5.1kΩ × 2)
Output voltage	3.3V ± 2% (regulated)
Output current (max)	800 mA continuous (datasheet AMS1117)
Dropout voltage	≤ 1.3V at 800mA
Quiescent current	≈ 5 mA (no load)
Power LED	On while regulated output present
Dimensions	50 × 30 mm
Layer count	2 layers, 1.6mm FR-4
Operating temperature	0 to +70 °C (commercial)

Absolute Maximum Ratings

Parameter	Min	Max	Unit
VBUS input	-0.3	6.5	V
+3V3 output (continuous)	—	800	mA
Storage temperature	-65	+150	°C
Operating temperature	0	+70	°C
Junction temperature	—	+125	°C

2.0 SCHEMATIC

Functional schematic in IPC-style hierarchical form. The overview shows every component and net on one sheet; each subsequent page is the detail of one named subcircuit (usbc_power, ldo, led, header, ...) — useful for reviewers who want to read the design top-down.

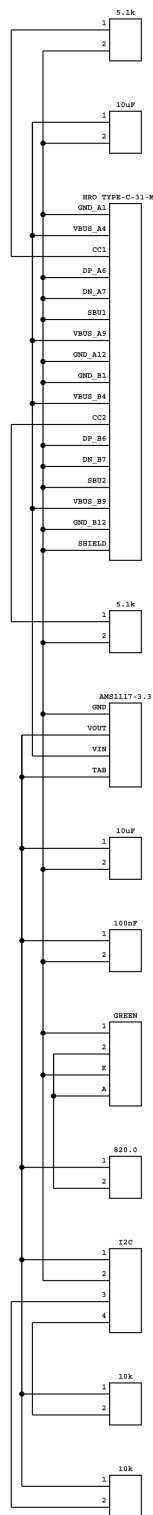


Figure 4. Overview — full flat schematic.

2.1 SCHEMATIC DETAIL — USBC_POWER

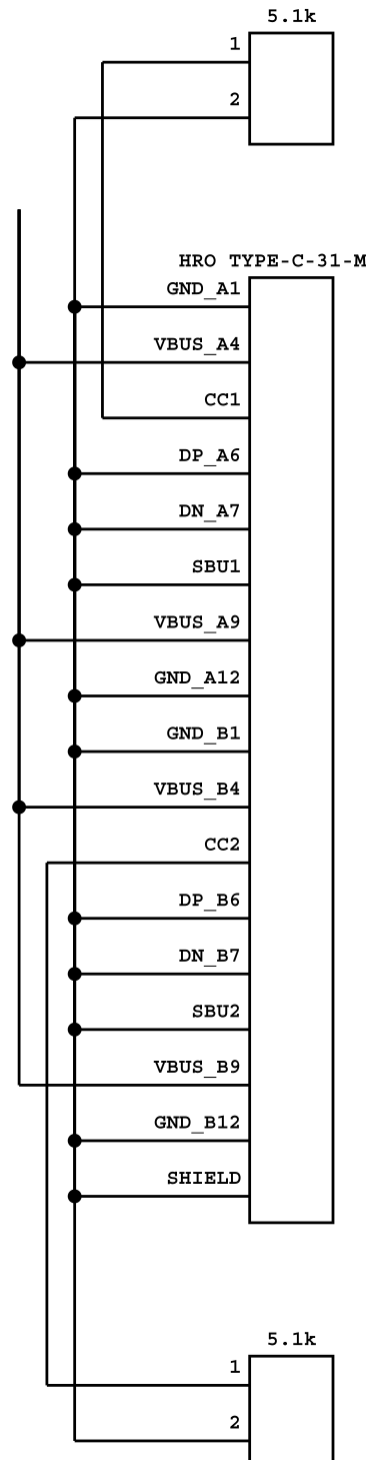


Figure 5. `usbc_power` subcircuit.

2.2 SCHEMATIC DETAIL — LDO

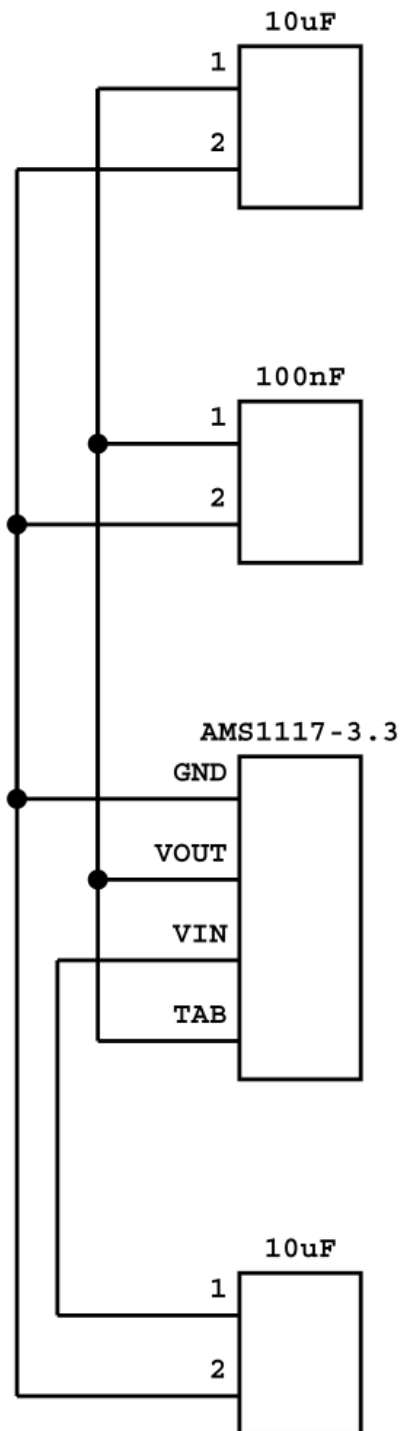


Figure 6. Ldo subcircuit.

2.3 SCHEMATIC DETAIL — I2C_PULLUPS

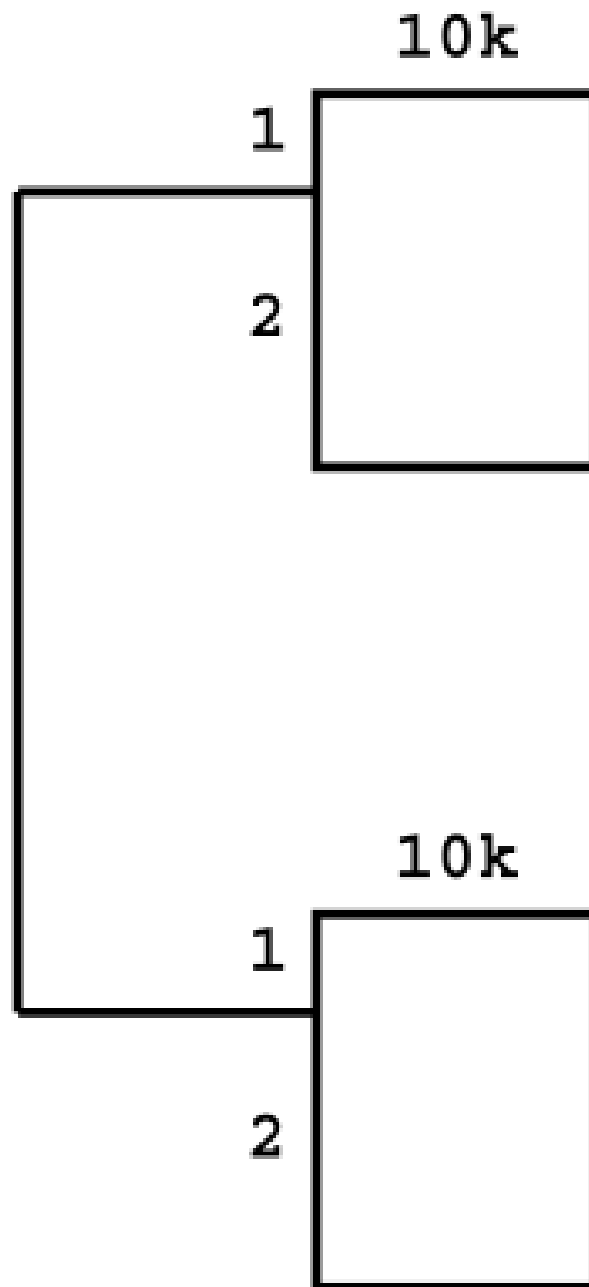


Figure 7. *i2c_pullups* subcircuit.

2.4 SCHEMATIC DETAIL — LED

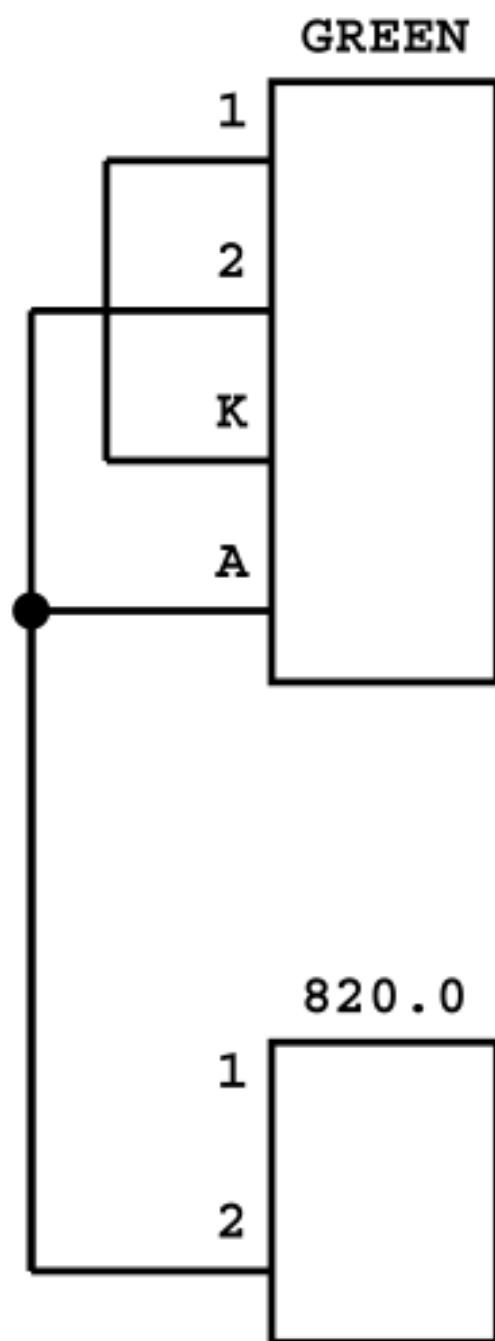


Figure 8. led subcircuit.

2.5 SCHEMATIC DETAIL — HEADER

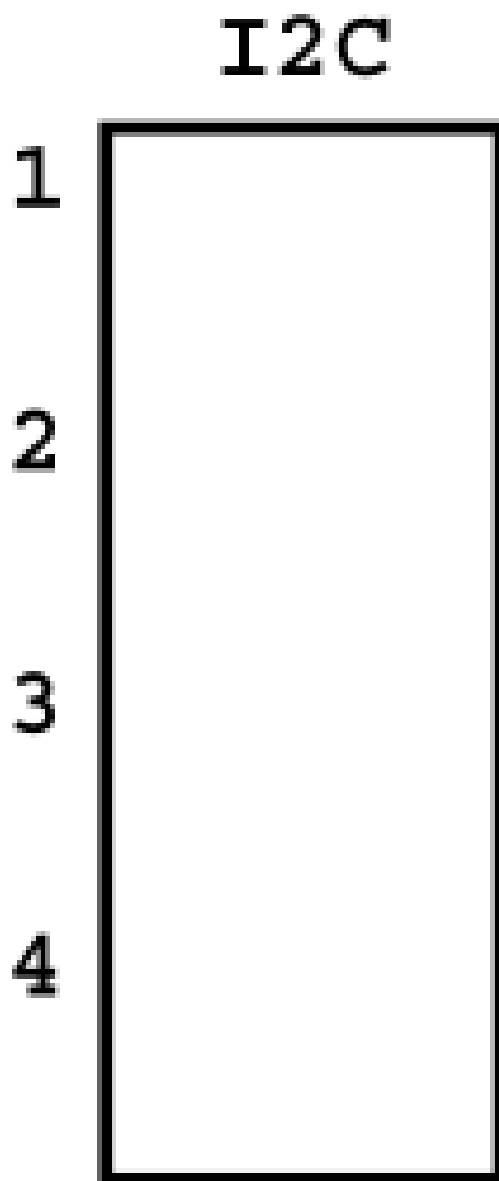


Figure 9. header subcircuit.

2.6 SCHEMATIC DETAIL — MOUNTING

(mounting embed failed: cannot write empty image)

3.0 PCB LAYOUT

Two-layer FR-4, 1.6 mm thickness, 50 × 30 mm. Top copper carries signal and power; bottom copper hosts crossover routes. GND copper pour fills both layers.

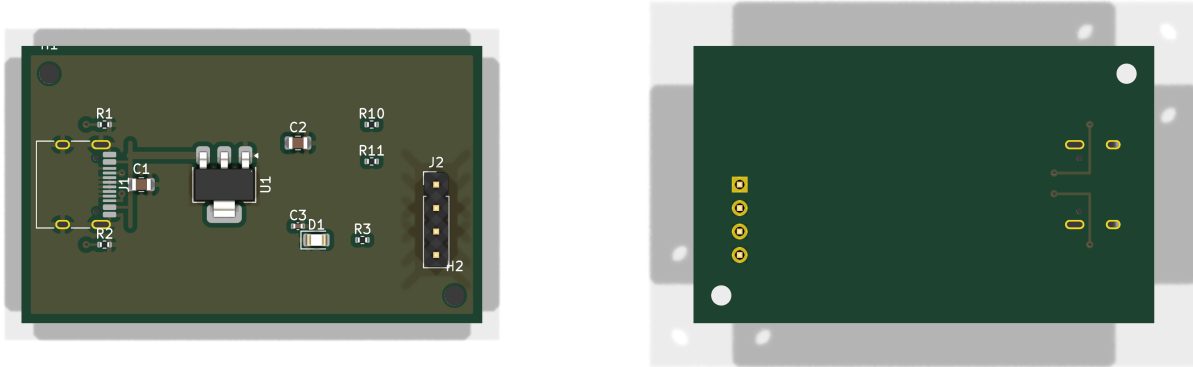


Figure 2. Top view. Figure 3. Bottom view.

3.1 PCB STYLIZED VIEW

Schematic-style 2D rendering generated by pcbdraw — copper tracks, silkscreen, board outline, and pad shapes in vector form. Useful for inspecting routing without the visual noise of the photoreal 3D view.

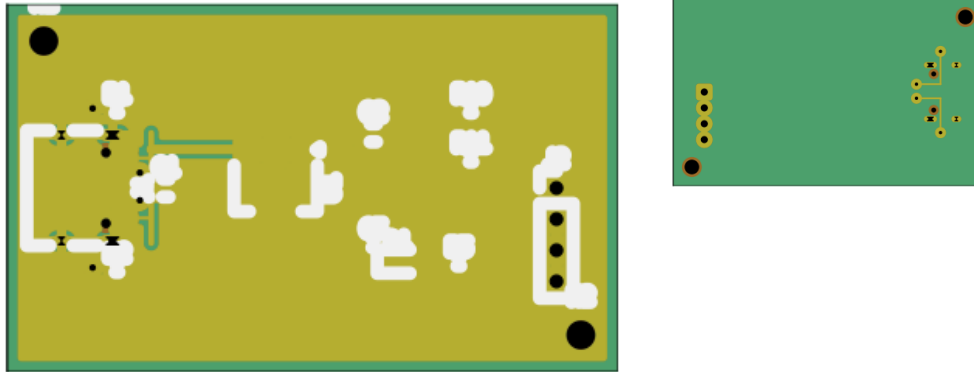


Figure 3a. Front (stylized). Figure 3b. Back (stylized).

4.0 CONNECTOR PIN DESCRIPTIONS

J1 — HRO TYPE-C-31-M-12

Pin	Net	Function
A1	GND	Ground reference
A12	GND	Ground reference
A4	VBUS	USB-C bus power input (5V nominal)
A5	CC1	USB-C Configuration Channel 1 (Rd to GND)
A6	GND	Ground reference
A7	GND	Ground reference
A8	GND	Ground reference
A9	VBUS	USB-C bus power input (5V nominal)
B1	GND	Ground reference
B12	GND	Ground reference
B4	VBUS	USB-C bus power input (5V nominal)
B5	CC2	USB-C Configuration Channel 2 (Rd to GND)
B6	GND	Ground reference
B7	GND	Ground reference
B8	GND	Ground reference
B9	VBUS	USB-C bus power input (5V nominal)
SH	GND	Ground reference

J2 — I2C

Pin	Net	Function
1	+3V3	Regulated 3.3V output
2	GND	Ground reference
3	SDA	Signal
4	SCL	Signal

5.0 BILL OF MATERIALS

All parts are JLCPCB-compatible. Where possible, basic-part LCSC numbers are used to avoid PCBA setup fees.

Designator	Qty	Value	Footprint	LCSC
C1, C2	2	10uF	C_0805_2012Metric	C15850
C3	1	100nF	C_0402_1005Metric	C1525
D1	1	GREEN	LED_0805_2012Metric	C72043
J1	1	HRO TYPE-C-31-M-12	USB_C_Receptacle_HRO_TYPE-C-31-M-12	C165948
J2	1	I2C	PinHeader_1x04_P2.54mm_Vertical	—
R1, R2	2	5.1k	R_0402_1005Metric	C25905
R10, R11	2	10k	R_0402_1005Metric	C25744
R3	1	820.0	R_0402_1005Metric	—
U1	1	AMS1117-3.3	SOT-223-3_TabPin2	C6186

Total: 12 components across 9 unique part numbers. Mounting hardware (H1–H4) is excluded from the assembly BOM.

6.0 BRINGUP PROCEDURE

(Bringup procedure not available — see bringup.md in source repo.)

7.0 SIMULATION RESULTS

Pre-fabrication SPICE simulation results. Each plot was generated by ngspice from the same netlist that produced the PCB — circuit and simulation share a single Python source.

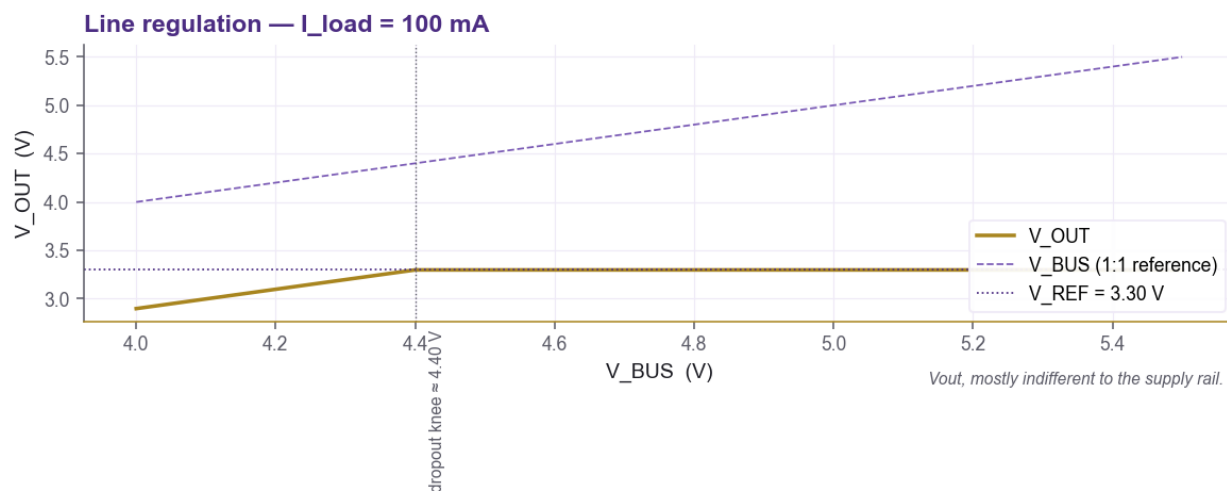
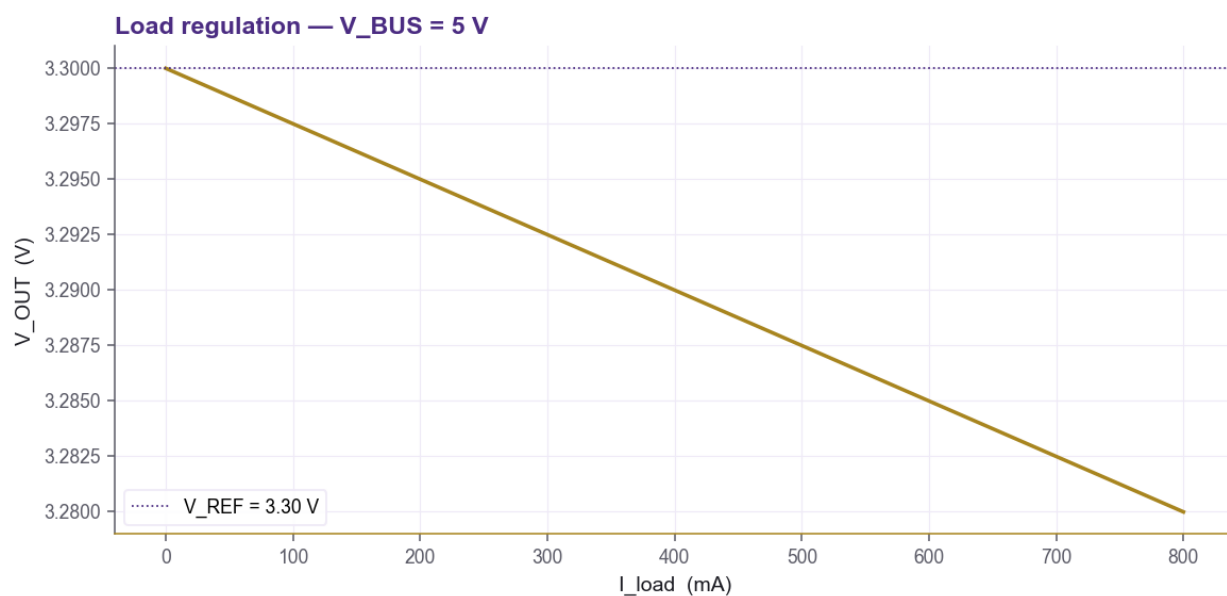
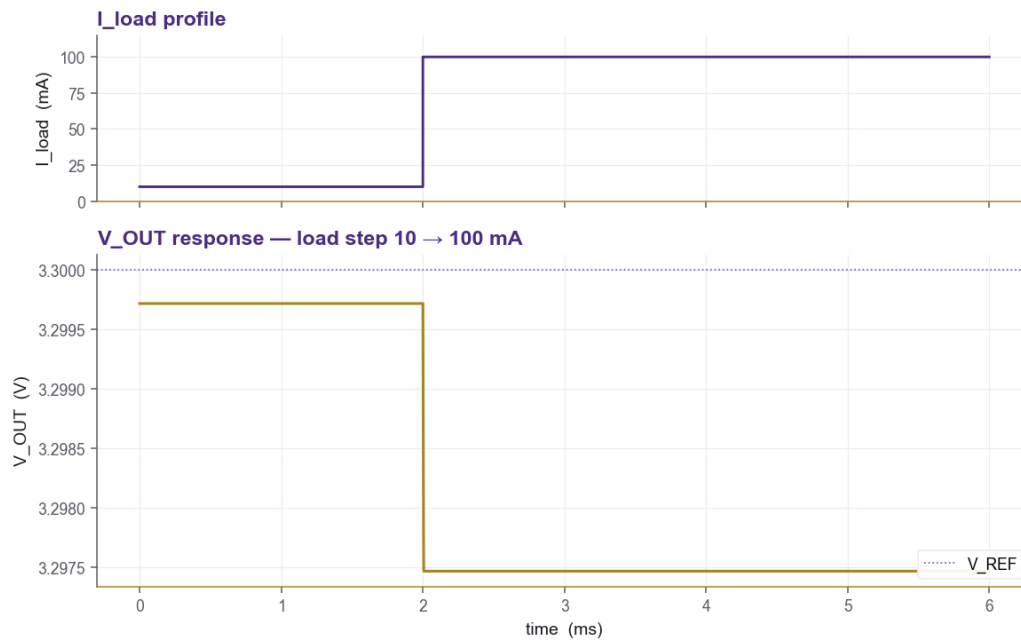


Figure 1. Line Reg.



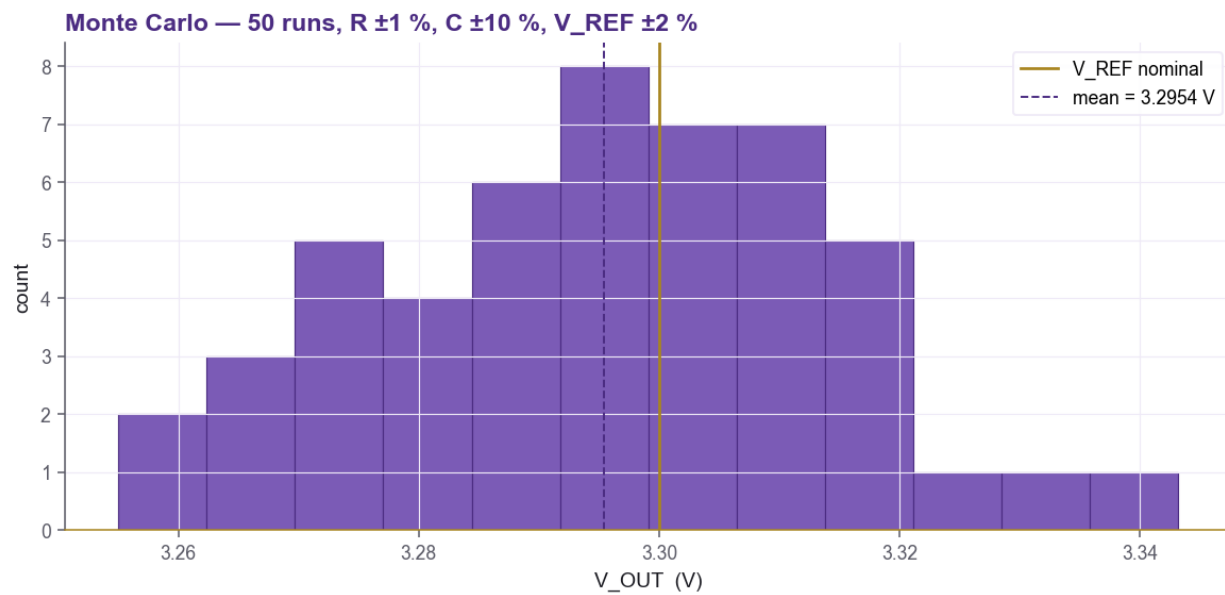
The price of current is, alas, paid in millivolts.

Figure 2. Load Reg.



A short pulse of disrespect, recovered with grace.

Figure 3. Load Step.



$\sigma = 18.76 \text{ mV}$ (0.569 %). Tolerances: tested, not feared.

Figure 4. Monte Carlo.

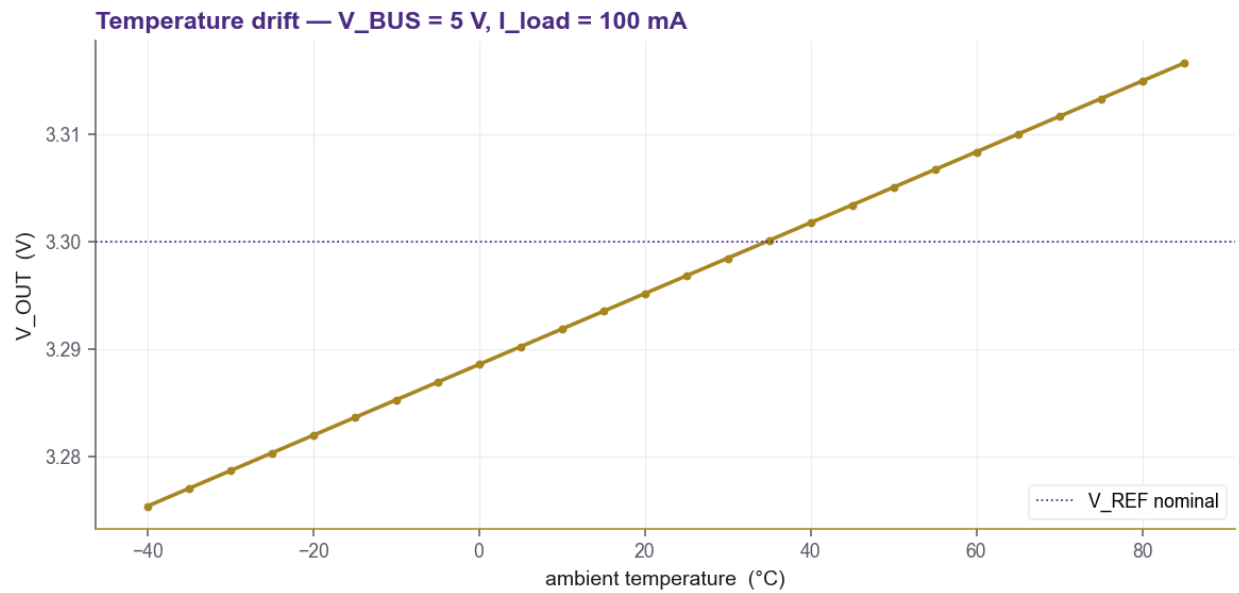
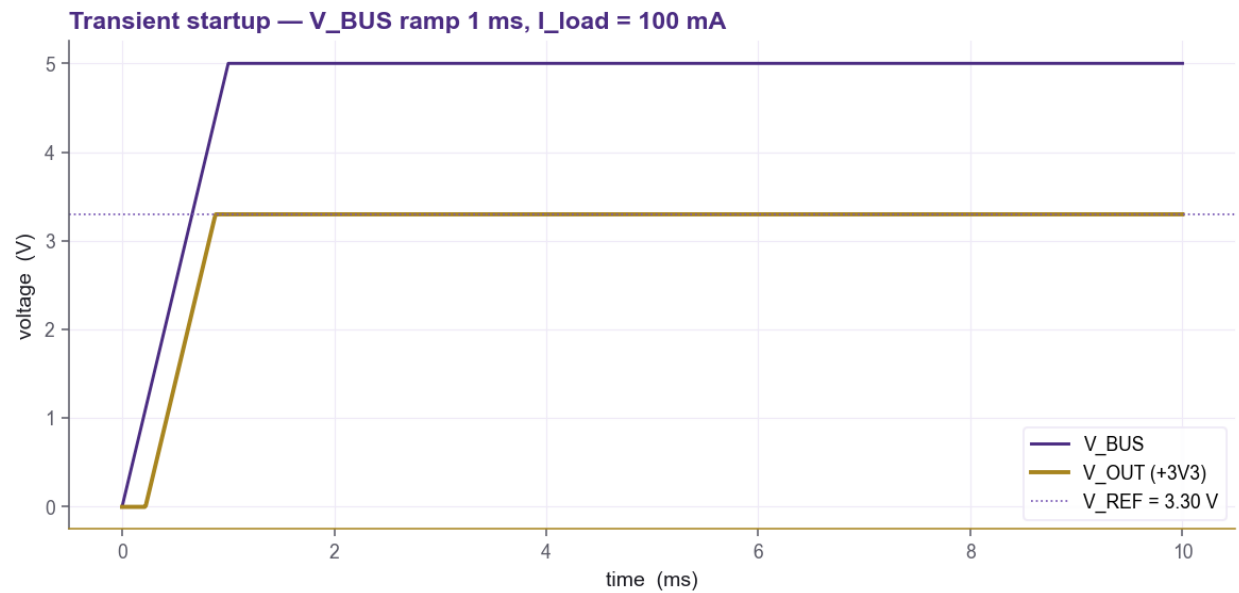


Figure 5. Temp Sweep.



Electrons reach steady state. Designers, eventually.

Figure 6. Transient.